

Public Works Transparency & Coordination Platform

Production-grade system architecture

Modular monolith, event-driven, microservice-ready. Designed for Indian municipal governance (ULB / utility / contractor / resident) and for several developers working in parallel.

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1. Reference systems and the key design lesson

Note: the notes below describe each reference from general knowledge. Login-gated portals (PM GatiShakti, DoT) and JS-heavy dashboards were not crawled live, so treat them as design inspiration.

Reference	What to borrow
PM GatiShakti	Layer-based geospatial coordination across departments; plans of every agency visible on one map before work starts.
DoT / Sanchar RoW portal	Single-window Right-of-Way permission for utilities with statutory timelines.
Vigeye GPMS (BBMP)	Geo-tagged, timestamped photo evidence for progress.
MCGM roads dashboard	Public contract-wise progress and status.
London Works / Street Manager	A registry where a permit is required before digging. The permit system comes first; the public map is a by-product.
Swachh.city	Complaint ticket with SLA timers and automatic escalation up the hierarchy.

Key design lesson. If the platform is only a display layer, agencies will not keep it updated. Make registration in the platform a precondition of the road-cut / RoW permission. Then the data stays fresh because the agency needs it to obtain permission. This is the single most important adoption lever.

2. Indian hierarchy mapped to platform roles

Actor	Typical chain / examples	Platform role and scope
Municipal Corporation	Commissioner, Chief Engineer, Superintending Engineer (zone), Executive Engineer (division), Assistant Engineer / JE (ward), site inspector	Staff roles scoped by jurisdiction: JE edits own ward, EE sees own division, CE/Commissioner see city.
Utility owners	Water and sewer board, DISCOM, gas distributor, telecom / fibre operators, metro rail, PWD / NHAI for roads they own	Agency-scoped editors. Can edit only their own works, but can see all geometries and dates (needed for conflict detection).
Other authorities	Traffic Police (closure NOC), third-party quality auditor, ward councillor, vigilance	Approver / auditor / read-only oversight roles.
Contractors	Excavation, laying, repair, restoration	Field role: submits progress and photos; JE verifies. Contractor identity is a public field where policy allows.
Residents	Citizens, RWAs, shopkeepers	OTP-verified login; follow works, submit project-linked feedback, see response history.

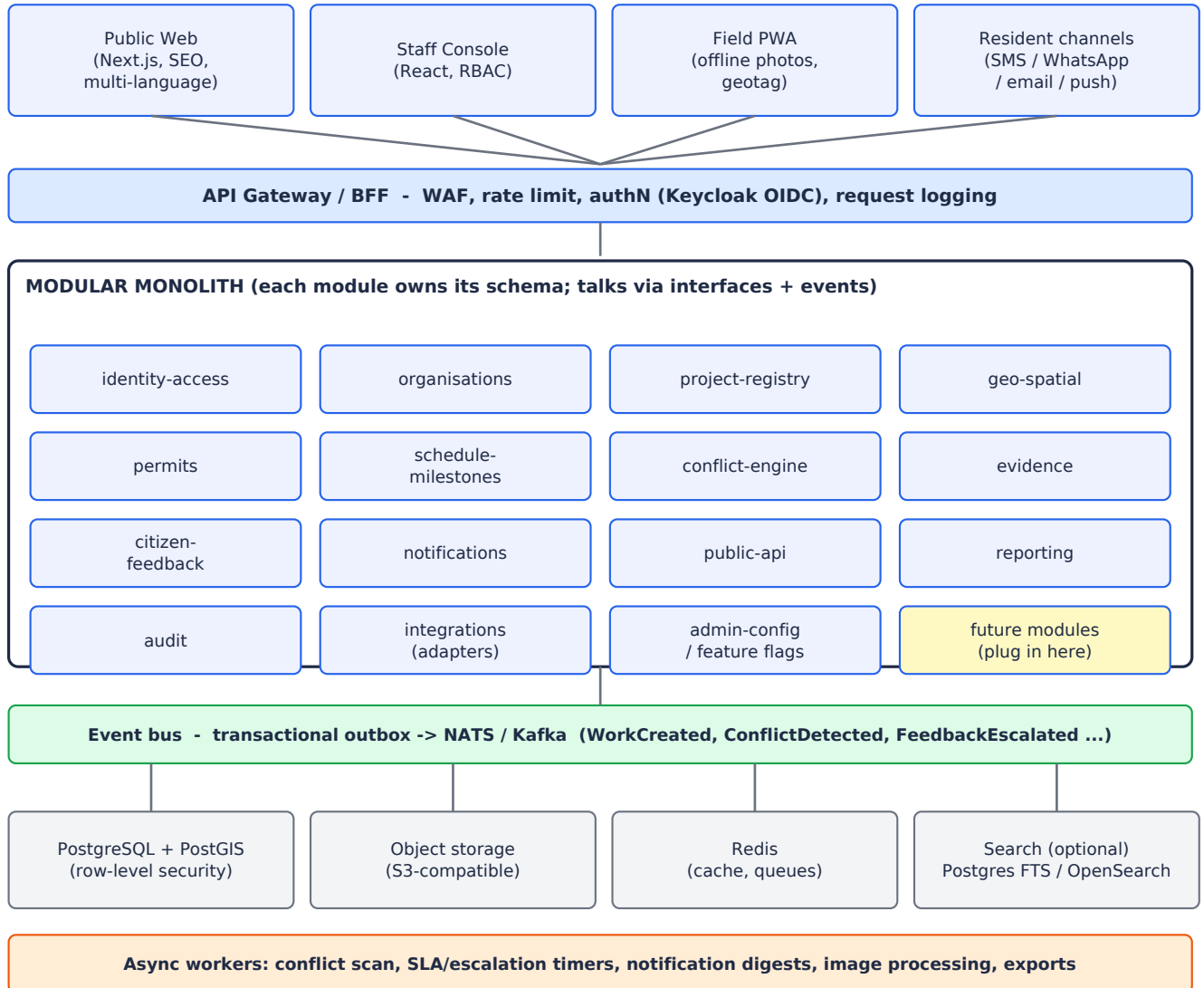
Grievance escalation ladder (configurable per city): JE → AE → EE → SE → CE → Commissioner → Mayor / Council. Outside-system channels (CM helpline, CPGRAMS, Lokayukta, RTI) are linked, not replaced.

Access model: Organisation → Role → Jurisdiction (ward / zone / division). RBAC for what you can do, ABAC for where you can do it. Every table carries *tenant_id* (city) with Postgres row-level security so one deployment can serve many cities later.

3. System architecture and technology choices

Style: modular monolith, microservice-ready. One deployable, one database cluster, simple debugging, and strict module boundaries so that any module can later be extracted into its own service without a rewrite.

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Layer	Recommended	Why
Backend	NestJS (TypeScript) or FastAPI / Django	Pick what the team knows; both support clean module boundaries.
Database	PostgreSQL + PostGIS	Geometry queries, date ranges, row-level security, JSONB for flexible fields.
Maps	MapLibre GL, OSM base tiles, vector tiles via Martin / pg_tileserv	Open, no per-view licence cost, fast road-segment rendering.
Public portal	Next.js	SEO, server rendering, fast on weak mobile networks.
Field app	Offline-capable PWA first (React Native later)	Geotagged photos with offline queue; no app-store dependency.
Identity	Keycloak (OIDC)	Staff SSO + MFA; resident mobile OTP.
Events	Transactional outbox → NATS JetStream or Kafka	No lost events; start light, scale later.
Files	S3-compatible object storage	Photos, attachments, exports.

Layer	Recommended	Why
Infra	Docker; Kubernetes or ECS in an Indian region; GitHub Actions	Data residency, repeatable deploys.
Observability	OpenTelemetry, Prometheus, Grafana, Sentry	Trace a request across modules and workers.

4. Modules and ownership

Rule: each module owns its tables (separate Postgres schema) and exposes only a public interface. **No module reads another module's tables directly.**

#	Module	Responsibility
1	identity-access	Users, roles, RBAC + jurisdiction scoping, resident OTP login.
2	organisations	Agencies, departments, zones / wards, officer directory, per-agency escalation ladder.
3	project-registry	The core <i>Work</i> entity: category, status state machine, contractor, public vs internal fields.
4	geo-spatial	Road network, wards, geometries, buffering, tiles. The only module doing geometry maths.
5	permits	Road-cut / RoW application, approval, restoration deposit, closure NOC. The adoption lever.
6	schedule-milestones	Planned dates, milestones, revisions with mandatory reason codes. History never overwritten.
7	conflict-engine	Detects overlap and recently-resurfaced cases; records the staff decision.
8	evidence	Dated, geotagged photos, EXIF checks, restoration-verified step.
9	citizen-feedback	Project-linked tickets, SLA timers, escalation, public / private split, moderation.
10	notifications	Follow project or area, digesting (no spam), channel adapters (SMS, WhatsApp, email, push).
11	public-api	Read-only open data (GeoJSON, CSV), cached.
12	reporting	Dashboards: counts by status / area / department, delayed, overdue updates, repeat-dig metrics.
13	audit	Append-only log of who changed what and when.
14	integrations	Adapters: GatiShakti layers, Sanchar RoW, municipal ERP / e-procurement, GPMS-style systems, CPGRAMS.
15	admin-config	Per-city settings: buffer metres, lookback months, SLA days, reason codes, feature flags.

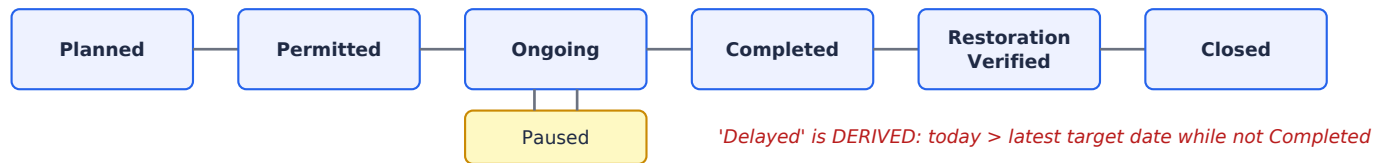
5. Module communication: commands and events

- **Commands** go through each module's typed service interface (synchronous).
- **Events** go through the bus using the **transactional outbox pattern**: the event row is written in the same DB transaction as the change, then a worker publishes it. This avoids 'saved but never announced'.
- Events are versioned (*WorkDatesRevised.v1*). New fields are additive, so old consumers never break.
- A new feature subscribes to existing events instead of editing old code. Example: a future 'traffic advisory' module just subscribes to *WorkStatusChanged* and *ClosureAdded*.

Event	Published by	Typical consumers
WorkCreated / WorkUpdated	project-registry	conflict-engine, audit, reporting, notifications
WorkDatesRevised	schedule-milestones	notifications, reporting, audit
WorkStatusChanged	project-registry	notifications, reporting, audit
ConflictDetected / ConflictResolved	conflict-engine	notifications (staff + residents), reporting, permits
PermitApproved / PermitHeld	permits	project-registry, notifications
FeedbackSubmitted / FeedbackEscalated	citizen-feedback	notifications, reporting, audit
EvidenceAdded / RestorationVerified	evidence	project-registry, notifications

6. Core designs

6.1 Work status state machine



A road cut is not truly finished at *Completed*; it is finished at *Restoration Verified*. Marking a work Paused, or changing a target date, requires a reason code and free-text explanation, and both the old and new dates stay visible publicly.

6.2 Conflict engine, step by step with numbers

Step	What happens	Numbers
Setup	Two works are registered with geometry and date range.	Work A (water board): 400 m trench on MG Road, 12 Jan - 20 Feb. Work B (telecom): 150 m duct on the same stretch, 10 Feb - 5 Mar.
Spatial test	Buffer each geometry (configurable) and test intersection with PostGIS.	Buffer = 15 m. ST_Intersects on buffered geometries = true.
Date test	Compare date ranges.	Overlap = 10 Feb to 20 Feb = 11 days. Raise Overlap alert.
Resurfacing test	Look back at completed resurfacing on the same segment.	Work C (electricity) proposed 1 Jun; road resurfaced 15 Mar = 78 days earlier. Lookback = 24 months, so raise Recently resurfaced .
Notify and decide	Both agencies' EEs get the alert with both projects, agencies and schedules.	Decision options: coordinate dates, combine work, reschedule, or justify both.
Outcome	The system never auto-cancels or auto-approves. It may hold a permit in 'pending coordination' until a decision is recorded.	Residents see a simple notice if nearby work is coordinated or rescheduled.

When it runs: synchronously on work / permit submission (instant warning to the submitter) and as a nightly scan for edits and bulk imports. Spatial index: GiST on geometry; date range index: GiST on daterange.

6.3 Citizen feedback and escalation (Swachh.city-style)

Time since submission	Action (default, configurable per city)
Day 0	Ticket created with reference number, linked to <i>work_id</i> ; routed to the work's responsible JE / contact.
Day 3 without response	Auto-escalate to Assistant Engineer; resident sees status change.
Day 7	Escalate to Executive Engineer; appears in staff 'awaiting response' view.
Day 14	Escalate to Superintending Engineer; counted in public dashboard as overdue.
Any time	Resident sees full status history. Private text and contact details stay hidden; only moderated items are published.

6.4 Data freshness and trust

- Overdue-update rule: no update in N days (for example 7) → public badge 'update overdue'.
- Explicit public / internal flag per field.
- Resident 'still ongoing?' confirmations act as a cross-check against stale agency data.
- Bulk onboarding: CSV / KML / Shapefile import so agencies do not retype existing work orders.

7. Core data model

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tenant (city)
└─ agency — officer/user — role_assignment (role, jurisdiction)
└─ work (title, purpose, category, status, geometry, agency_id, contractor_id, public_flags)
    └─ work_milestone (name, planned_date, actual_date, pct)
    └─ work_date_revision (old_target, new_target, reason_code, explanation, by_user, at)
    └─ work_status_history (from, to, reason, by_user, at)
    └─ evidence (file_ref, taken_at, lat/lon, verified_by)
    └─ permit (type, applied_at, approved_by, deposit, closure_noc)
    └─ conflict_alert (work_a, work_b, type, overlap_days, detected_at)
        └─ conflict_decision (action, note, decided_by, at)
    └─ feedback_ticket (ref_no, text, photo, location, status, is_public)
        └─ ticket_event (status, responder, message, at)
    └─ follow_subscription (resident_id, work_id | area_id, channel)
outbox_event (id, type, version, payload, created_at, published_at)
audit_log (actor, action, entity, before, after, at, prev_hash)

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8. Cross-cutting production concerns

Area	Decisions
Security	WAF, rate limiting, CAPTCHA on resident forms, strip EXIF GPS from public photos, field-level encryption for resident phone numbers, least-privilege DB roles, dependency scanning.
Compliance	DPDP Act 2023 (consent, purpose limitation, deletion), GIGW / WCAG accessibility, data hosted in India.
Reliability	Postgres backups with point-in-time recovery, read replica for the public portal, CDN and heavy caching for public GET endpoints and map tiles, health checks and autoscaling.
Audit integrity	Append-only audit log with hash chaining so tampering is detectable.
India realities	SMS needs DLT registration; WhatsApp Business API; English + Hindi + local language; low-bandwidth mode; offline field PWA.
Observability	Structured logs with correlation IDs, traces across modules and workers, alerting on queue lag and failed SLA timers.
Extensibility	Feature flags, versioned APIs (/v1), webhooks for external systems, adapter interface for new data sources, per-city config instead of code forks.

9. Team workflow for parallel development

- **Monorepo** with one folder per module and CODEOWNERS so each person owns a module.
- **Contract-first:** write the OpenAPI spec and event schemas before code. Frontend works against mocks.
- **Boundaries enforced in CI** with import-lint rules (for example dependency-cruiser) so a module cannot import another module's internals.
- **Own your migrations:** each module has its own schema and migration folder, so no merge fights on a shared migration file.
- **Contract tests** on events and public interfaces; a breaking change must bump the version.
- **Shared seed dataset:** a fake city with about 30 works, 3 agencies, a few planted conflicts, so everyone develops against the same data.
- **Shared kernel kept tiny:** only IDs, date / money types and the auth context.
- **Trunk-based development** with short-lived branches, feature flags for unfinished work, and per-module CI so one team's failure does not block the others.

Suggested ownership split for a team of four

Owner	Modules
Dev 1 - Platform	identity-access, organisations, admin-config, audit, infra / CI
Dev 2 - Core and geo	project-registry, geo-spatial, schedule-milestones, conflict-engine
Dev 3 - Citizen side	public web, public-api, citizen-feedback, notifications
Dev 4 - Field and integrations	staff console, field PWA, evidence, permits, integrations, reporting

10. Feature-to-module traceability

Requirement	Module(s)
Public works map and list, filters, last-updated date	public-api, geo-spatial, project-registry, reporting
Project detail page	project-registry, schedule-milestones, evidence, organisations
Department / agency submission and update form, who-changed-what	project-registry, identity-access, audit, permits
Overlap and conflict detection, staff decision, resident notice	conflict-engine, notifications, permits
Progress and delay history, mandatory delay reason, overdue flags	schedule-milestones, project-registry, reporting
Resident feedback tied to a project, reference number, status history	citizen-feedback, notifications
Notifications (follow project / area, no spam)	notifications
Accountability dashboard, repeat-work metrics, staff unresolved view	reporting, conflict-engine, citizen-feedback
Escalation up the hierarchy	organisations, citizen-feedback

11. Build order

Phase	Deliverable
1 - Spine	identity-access, organisations, project-registry, geo-spatial, audit
2 - Visibility	Public map and project detail, agency submission form, bulk import
3 - Honest dates	Schedule revisions with reasons, delay and overdue flags
4 - Coordination	conflict-engine and permits (registration required for road-cut permission)
5 - Accountability	citizen-feedback with SLA escalation, then notifications

Phase	Deliverable
6 - Scale	reporting dashboards, integrations (GatiShakti, RoW, ERP), multi-city

Out of scope for version 1: payroll, tender bidding, contractor payments, full asset management, traffic simulation, AI completion prediction, a general-purpose complaint system. They add scope without solving the core problem.